Wave speed

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use speed, frequency and wavelength in a consistent relationship.

SUCCESS CRITERIA - I can...

1. Use v=fλ with matching units.
2. Check frequency and wavelength changes at fixed speed.
3. Distinguish medium speed from oscillation rate.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A system absorbs 70 J heat and does 30 J work. Find its internal-energy change using $\Delta U=Q-W_{by}$.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: v = fλ; λ = v/f.
At fixed wave speed v, increasing frequency f decreases wavelength λ.

Worked reasoning: A wave with wavelength 3 m and frequency 2 Hz travels at v=fλ=6 m/s. At the same speed, frequency 3 Hz would require wavelength 2 m.

Your turn: Draw both waves over the same 6 m length and label their frequencies.

Marzano check - circle: ① ② ③

03 Look and explain

Two waves travel in the same ideal medium at the same speed. One has twice the frequency. Compare wavelengths.

Marzano check - circle: ① ② ③

04 Build the solution

A wave has f=4 Hz andλ=2 m. Calculate v. Another wave has f=8 Hz at that same speed; findλ=v÷f.

Marzano check - circle: ① ② ③

05 Use the idea

A student writes v=f/λ. Check units and explain why that is not a speed.

Marzano check - circle: ① ② ③

06 Choose a method

Sunlight carries energy across 4 million km of nearly empty space in a simplified example. Identify the transfer mechanism and explain why matter is unnecessary.

Marzano check - circle: ① ② ③

07 Exit check

f=3 Hz,λ=5 m. Find v.

Marzano check - circle: ① ② ③

08 STRETCH: Apply it another way

Create three frequency-wavelength pairs that all give 12 m/s. Sketch one pair.

Marzano check - circle: ① ② ③